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Topic: Mini Project

Project Title

Smart Route Planner AI System Using BFS, UCS, and A Search Algorithms*

Introduction

The Smart Route Planner AI System is an Artificial Intelligence project developed using Python. The main purpose of this project is to find the best route between two cities by using different AI search algorithms. The project compares the performance of Breadth-First Search (BFS), Uniform Cost Search (UCS), and *A Search** to determine the most efficient path.

Objective

The objectives of this project are:

- To build a Smart Route Planner using Python.
- To implement BFS, UCS, and A* search algorithms.
- To compare the performance of different search algorithms.
- To find the shortest and most efficient route between cities.

- To visualize the city network and algorithm performance using graphs.

Description

In this project, cities are connected in the form of a graph where each city is a node and each road has a distance value. The user enters a starting city and a destination city. The system then applies BFS, UCS, and A* algorithms to find the route.

After running all three algorithms, the project compares their results based on:

- Route found
- Total cost
- Number of nodes explored

The project also includes graph visualizations to show the city network and compare the performance of the algorithms.

Tools and Technologies Used

- Python
- Jupyter Notebook (Anaconda)
- NetworkX
- Matplotlib
- Heapq
- Collections (Deque)

Algorithms Used

Breadth-First Search (BFS)

BFS searches the graph level by level. It finds a path between two cities but does not consider the travel cost.

Uniform Cost Search (UCS)

UCS finds the route with the lowest total cost by considering the distance between cities.

*A Search**

A* Search uses both the travel cost and heuristic values to find the shortest and most efficient route.

Features

- User-friendly interface
- City graph representation
- Route finding between cities
- Cost calculation
- Algorithm comparison
- Performance analysis
- Graph visualization

Results

The project successfully finds routes between cities using BFS, UCS, and A* algorithms. The comparison shows that A* finds the best route more efficiently by exploring fewer nodes while still providing the optimal path.

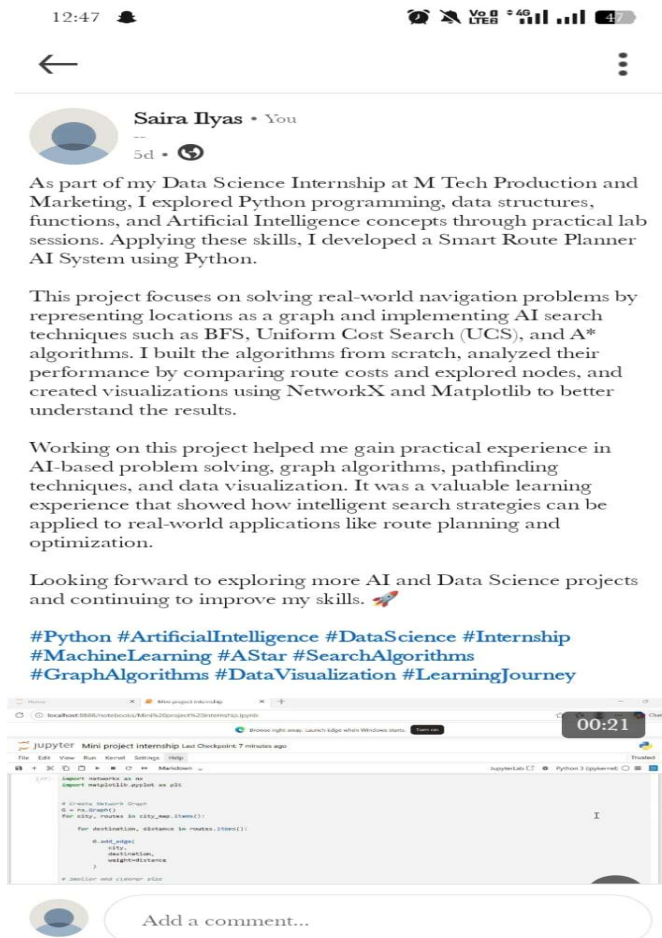
Applications

- GPS Navigation
- Route Planning
- Delivery Services
- Transportation Systems
- Robot Navigation
- Smart City Applications

Conclusion

The Smart Route Planner AI System demonstrates how Artificial Intelligence search algorithms can solve real-world route planning problems. By comparing BFS, UCS, and A* algorithms, the project shows the importance of selecting the right algorithm for efficient pathfinding. This project also improved my understanding of Python programming, graph data structures, AI search techniques, and data visualization.

Project Sharing on LinkedIn



After completing the **Smart Route Planner AI System**, I shared the project on my LinkedIn profile to showcase my learning and practical work completed during my Data Science Internship. The post highlights the project's objectives, the AI search algorithms used (BFS, UCS, and A*), and the graph visualizations created in Python.

Sharing the project on LinkedIn helped me document my work, build my professional portfolio, and connect with other students and professionals in the fields of Artificial Intelligence and Data Science.